Welcome, please sit in a group of 5-6. Pull over a chair if needed.

Analog Theremin

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What is a theremin?

Objective and Goals

- **Concept** : A fully analog theremin controlled by hand proximity or other measures.
- **Deliverable** : Functional theremin unit with pitch & volume.
- **Team Size** : 4-5 students per Theremin
- **Use Case** : Demo instrument for interactive showcase

This is not even Martha Stewart

What concepts will you develop through this project?

Technical: LTSpice, Oscilloscope, Signal Generator, Antennas, Op Amps, TFs

Theoretical: Circuit Design, Passive/Active Filter Design, Analog Circuits

Other: Lab Test, Teamwork, Integration

Timeline

Weeks 1-2

Understanding the Basics

Review DC and AC Circuits as well as Laplace/Fourier Transforms

Form teams

Begin design process

Weeks 3-5

Prototyping

Begin to tinker with materials and get something to work

Weeks 6-7

Construction

Begin bringing pieces together and construct a working prototype.

Week 7

Detailed Requirements Drafting

Time to document specifically what the project is going to require

Week 8-9

Development

Design, validate, build, verify

Week 10

Document and Present

Document lessons learned, accomplishments, and share your discoveries with The Boring Club and the UW community

Kahoot

Go to kahoot.it, and wait for the code.

Kahoot Link

Group activity

Come up with at least one question for each topic that nobody in your group can answer:

- Laplace Transform
- Fourier Transform
- Circuit Design (Kirchoff’s Laws, Ohm’s Law)
- Filter Design (Capacitors, Inductors, Op-Amps)
- Lab Equipment (Signal Generator, Oscilloscope)

Fourier Transform

Purpose: Find the frequency content of a signal

Questions:

Laplace Transform

Purpose: Linearize differential equations and circuits

Questions:

Circuit Design

Purpose: Analyze component behavior to create a model before building it

Questions:

Filter Design

Purpose: Control the pitch and intensity of the sound

Questions: How do we filter frequencies in an active filter? - Capacitors

Lab Equipment

Purpose: Test our designs. Theory can only take you so far.

Questions:

Concept Familiarity

Analog Theremin: Key Concepts

Next week:

Groups will begin brainstorming

Create a high-level diagram for your project

Write requirements for your design

Schematic creation

Every theremin will be different! There is not a single correct solution.

Thanks for watching

BE BOUNDLESS